

Sustainable Crop Management Systems

Land Sciences

May, 2026

Sustainable Crop Management Systems (A11592)

Short Title: Sustainable Crop Mgmt Systems
Faculty Science and Computing
Department Land Sciences
Cluster Agriculture
Author JGERAGHTY
Credits 5

Level: Advanced

Description of Module / Aims

This module is designed for students who already have an intermediate knowledge of crop production. The course will start by looking at plant physiology and factors affecting plant growth. It will then look at crop nutrition and the role of macro-nutrients and micro-nutrients in the plant. The role of alternative crops will be explored, including their end use. The management and operation of sustainable cropping systems will be examined and compared with intensive systems. New developments in plant breeding and biotechnology will also be covered.

Programmes

AGRI -0109	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	4 / 7 / M
AGRI -0109	BSc (Hons) in Land Management (in Agriculture) (WD_SBSAG_B)	1 / 1 / M

Indicative Content

- Plant physiology and plant growth
- Crop nutrition for sustainable production
- Production of alternative crops and their end uses
- Sustainable cropping systems, no-tillage and conservation agriculture
- Conventional plant breeding and genetically enhanced crop technologies

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Establish the interaction between plant physiology and factors affecting plant growth.
2. Determine appropriate crop nutrition including macro and micro nutrients to optimise sustainable crop growth and development.
3. Evaluate the role of alternative crops and their uses in modern agriculture.
4. Compare the management of conventional intensive systems with sustainable cropping systems.
5. Critique developments in conventional plant breeding systems and biotechnology systems in crop production and development.
6. Prepare a critical analysis of a published research journal paper.

Learning and Teaching Methods

- Lectures.
- Outdoor demonstrations.
- Expert lecturers.
- Webcast presentations.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Group Project	30%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Unable to analyse key topics in the course.

40%-49%: Demonstrate a good knowledge of the key learning outcomes of the course with basic evaluation of aspects.

50%-59%: Demonstrate a good level of knowledge with the aptitude to integrate and critically evaluate the various components of the course.

60%-69%: Demonstrate more detailed knowledge of the key topics, including critical evaluation of the roles and impacts of different crop production methods.

70%-100%: Demonstrate a comprehensive knowledge and understanding of all learning outcomes, including the ability to integrate and critically analyse the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- Alexander, S., A. Black, A. Boland, J. Burke, O.T. Carton and Coulter, B.S., Culleton, N., Dillon, P., Hackett, R., Humphreys, J., Keady, T., Lalor, S., Mchoul, J., Merfield, C., Murphy, B., O'Kiely, P., O'Riordan, E., Orlovius, K., Plunkett, M. & Tunney, H. *Major and Micro Nutrient Advice for Productive Agricultural Crops*. 3rd ed. Johnstown Castle Research Centre: Teagasc, 2008.
- Lammerts van Beuren, E.T. and J.R. Myers. *Organic Crop Breeding*. 1st ed. New Jersey: Wiley-Blackwell, 2012.
- Newell McGloughlin, M. and J.I. Burke. *Biotechnology - Present Position and Future Developments*. 1st ed. Dublin: Teagasc, 2000.
- Taiz, L. and E. Zeiger. *Plant Physiology*. 5th ed. Massachusetts: Sinauer Associates, Inc., 2010.
- Wolf, B. and G.H. Snyder. *Sustainable Soils - The Place of Organic Matter in Sustaining Soils and their Productivity*. 1st ed. New York: The Haworth Press, 2003.

Supplementary Material

- BFS, Anon. *BFS - Nutrient Deficiency Identification Guide*. Essex, UK.: Billericay Fertiliser Services Ltd, 2010.